

Membrane-Proximal mRNA Translation as a Strategy for Functional Membrane Protein Integration in Synthetic Cells

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Abstract

A central challenge in constructing functional synthetic cells is the de novo synthesis and proper integration of hydrophobic membrane proteins, which are indispensable for transport, signaling, and energy conversion. Natural cells overcome this obstacle by coupling mRNA localization with co-translational membrane insertion, but such spatial regulation has been largely absent from bottom-up synthetic systems. Here, we present a membrane-proximal translation platform that anchors mRNAs at lipid bilayers via programmable hybridization, thereby recruiting ribosomes for localized protein synthesis directly at the membrane interface. This spatially controlled process ensures correct topology and functional integration of diverse classes of membrane proteins, including single-pass and multi-pass transmembrane proteins, as well as higher-order oligomeric assemblies. Importantly, by encoding spatial information into the untranslated regions of mRNAs, we achieve programmable stoichiometric control over protein expression, allowing rational design of membrane composition in synthetic cells. Functional validation using the multidrug transporter EmrE demonstrates selective substrate uptake, confirming that the synthesized proteins are not only integrated but also active. Together, these results establish a minimal yet versatile framework that connects spatial mRNA regulation with membrane protein biogenesis. Our strategy provides a broadly applicable route for endowing synthetic cells with membrane-based functions, advancing their potential as platforms for biosensing, artificial signaling networks, and engineered organelles.